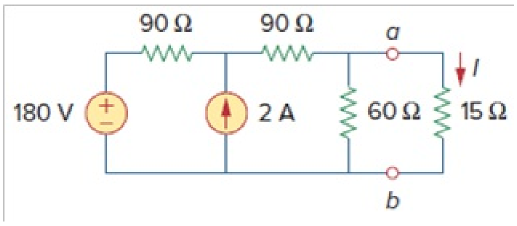
Chapter 4, Problem 8PP

Problem

Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4.30. Then find *I*.

Figure. 4.30:



Step-by-step solution

Step 1 of 7

Refer to Figure 4.30 in the text book.

Calculate the value of Thevenin's resistance.

Remove the load resistance and deactivate the sources from the circuit diagram as shown in Figure 1.

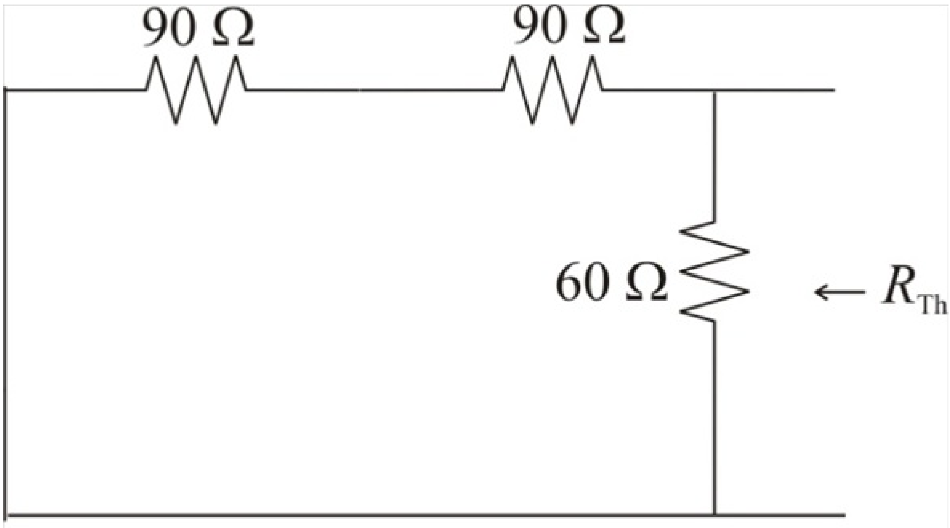


Figure 1

Step 2 of 7

Calculate the value of the Thevenin's resistance, R_{Th}.

$$\begin{aligned} R_{Th} &= 60 \| (90+90) \\ &= 60 \| 180 \\ &= \frac{(60)(180)}{60+180} \\ &= \frac{(60)(180)}{240} \end{aligned}$$

$$\begin{aligned} R_{Th} &= \frac{180}{4} \\ &= 45\ \Omega \end{aligned}$$

Therefore, the value of the Thevenin's resistance R_{Th} is $\boxed{45\ \Omega}$.

Step 3 of 7

Redraw the circuit diagram with the sources as shown in Figure 2.

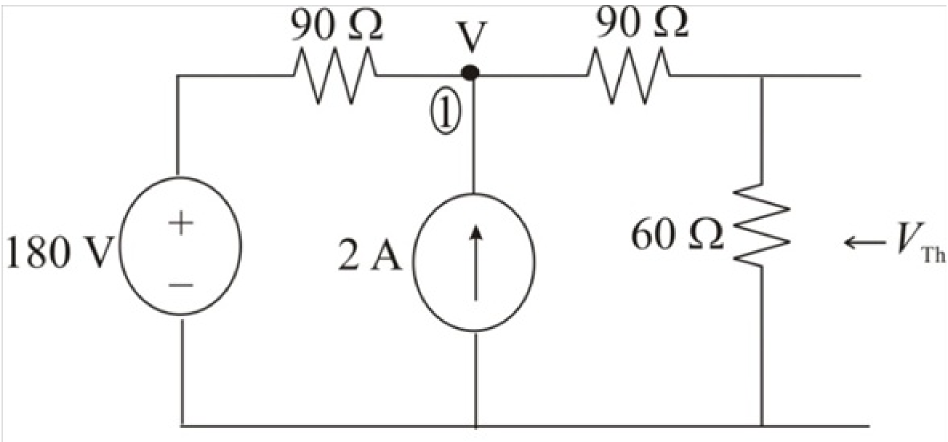


Figure 2

Step 4 of 7

Apply Kirchhoff's Current Law at node 1.

$$\frac{V-180}{90} - 2 + \frac{V}{90+60} = 0$$

$$\frac{V-180}{90} - 2 + \frac{V}{150} = 0$$

$$V\left[\frac{1}{90} + \frac{1}{150}\right] = 2 + \frac{180}{90}$$

$$V\left[\frac{4}{225}\right] = 4$$

$$\begin{aligned} V &= \frac{(225)(4)}{4} \\ &= 225\ V \end{aligned}$$

Therefore, the value of the voltage V is $225\ V$.

Step 5 of 7

Calculate the value of the Thvenin’s voltage, V_{Th}.

$$\begin{aligned} V_{\mathrm{Th}} &= (\mathrm{V})\left(\frac{60}{60+90}\right) \\ &= (\mathrm{V})\left(\frac{60}{150}\right) \\ &= (\mathrm{V})\left(\frac{2}{5}\right) \end{aligned}$$

Substitute $225\ \mathrm{V}$ for V in the equation.

$$\begin{aligned} V_{\mathrm{Th}} &= (225)\left(\frac{2}{5}\right) \\ &= (45)(2) \\ &= 90\ \mathrm{V} \end{aligned}$$

Therefore, the value of the Thevenin’s Voltage V_{Th} is $\boxed{90\ \mathrm{V}}$.

Step 6 of 7

Draw the Thevenin’s equivalent circuit diagram including load resistance.

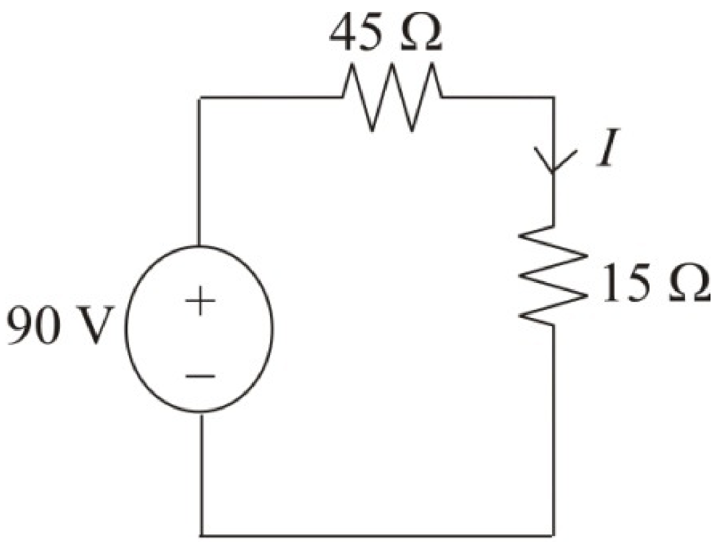


Figure 3

Step 7 of 7

Calculate the value of the current, I.

$$\begin{aligned} I &= \frac{90}{45+15} \\ &= \frac{90}{60} \\ &= 1.5\ \mathrm{A} \end{aligned}$$

Therefore, the value of the current I is $\boxed{1.5\ \mathrm{A}}$.